

Sam Belliveau

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EDUCATION

Cornell University

B.S. in Electrical & Computer Engineering

Ithaca, NY

Expected May 2026

- **GPA:** 3.7 / 4.3
- **Coursework:** Digital Systems Design (Microcontrollers), High-Level Synthesis, Signals & Systems, Introduction to Algorithms, Reinforcement Learning, Computer Architecture.

Hofstra University

B.S. in Computer Engineering (Transferred)

Hempstead, NY

Aug 2022 – May 2023

- **GPA:** 3.83 / 4.0

PUBLICATIONS & MANUSCRIPTS

CaptureGraph: A Distribution-Aware Programmable Capture Framework

In Prep (SIGGRAPH 2026)

S. Belliveau, A. Davis

Contribution: First Author. Architected a Python-based DSL and iOS runtime to decouple capture procedures.

CineCraft: Using Shot Plans for Cinematic Capture Guidance

Under Review (CHI 2026)

N. Tran, S. Belliveau, C. Xu, A. Davis

Contribution: Second Author. Engineered real-time subject tracking loop and stabilization filter.

Dynabox: Digitizing Analog Talkbox Performances via ARX Modeling

In Prep (UIST 2026)

J. Klein, S. Belliveau, A. Davis

Contribution: Co-Author. Implemented ARX models to isolate vocal tract resonances.

RESEARCH EXPERIENCE

Teaching Assistant – ECE 4760: Microcontrollers

January 2025 – Present

Cornell University

Ithaca, NY

- Graded laboratory assignments and provided feedback to students on reports and code quality
- Assisted student groups during lab sessions with debugging hardware/software and completing experiments
- Documented pin configurations and hardware specifications for laboratory equipment to improve student workflow

Cornell University (Prof. Abe Davis Lab)

Ithaca, NY

Undergraduate Researcher

June 2023 – Present

- **CaptureGraph (System Architect):** Developed a domain-specific language (DSL) that serializes experimental procedures into a graph data model. Built a companion iOS app that interprets this graph to provide state-aware AR guidance and real-time data validation, eliminating the need for hard-coded capture apps.
- **ReCapture (Modernization):** Refactored 7,000+ lines of legacy Swift code to enable HDR and RAW photography. Identified the “rigidity” bottleneck in bespoke apps which motivated the pivot to the programmable CaptureGraph framework.
- **CineCraft (Mobile Vision):** Addressed a critical HCI/Signal-Processing challenge where subject tracking fought with zoom automation. Implemented a stabilization filter using Apple’s Vision framework to enable smooth “virtual focus puller” capabilities.

PROJECTS

Real-Time Sound Localization System | *Embedded Systems (C, RP2040)*

Spring 2025

- Architected a < \$20 acoustic camera on a Raspberry Pi Pico. Achieved 50kHz sampling across three microphones using a custom ‘ping-pong’ DMA buffer strategy to ensure zero-latency processing.
- Interleaved a lightweight linear comb filter for event detection with a heavy cross-correlation engine for TDOA (Time-Difference-of-Arrival) estimation.

SmolVLA: FPGA Accelerator for Vision-Language-Action Models | *High-Level Synthesis (C++, Python)*

Fall 2025

- Designed a spatial architecture on Xilinx Alveo U280 to accelerate the Vision Encoder of the SmolVLA model.
- Implemented a ‘streaming Softmax’ pipeline to overlap QKV projection with score calculation, achieving ~20ms latency and ~277 MACs/Byte operational intensity.